Fiscal Year:	FY 2023	**Task Last Updated:**	FY 12/26/2024
PI Name:	Buettmann, Evan Ph.D.		
Project Title:	Investigating the Effects of Simulated Microgravity Duration and Connexin 43 Deficiency on Bone Fracture Healing		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	23284-9097	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020 TRISH-RFA-2001-PD: Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
Start Date:	09/01/2020	**End Date:**	08/31/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 08/31/2023 per TRISH (Ed., 8/4/22).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Donahue, Henry Ph.D. (MENTOR: Virginia Commonwealth University)		
Grant/Contract No.:	NNX16AO69A-P0501		
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Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP Astronauts exposed to long periods of unloading due to extended spaceflight experience on average a decrease in bone strength at 2.0 - 2.5% per month. This sharp decline in bone strength can predispose astronauts to fragility fractures, especially when re-entering a gravity-based loading environment due to extra-vehicular activities, extraterrestrial exploration, off-nominal spacecraft landings, and or upon return to Earth. While emerging evidence suggests that unloading, as would occur in microgravity during spaceflight, impairs fracture healing, the cellular and molecular mechanisms by which this occurs largely remains unknown due to a lack of ground-based rodent analog models mimicking spaceflight conditions. Understanding the mechanisms underlying the microgravity-induced impairment in bone regeneration following fracture will lead to the development of new countermeasure targets. One potential countermeasure target is Connexin 43 (Cx43), the primary gap junction protein in bone. Gap junctions facilitate intercellular communication between neighboring bone cells such as osteoblasts and osteocytes and have been strongly implicated in bone fracture healing and bone adaptation to the mechanical environment. In order to study how the duration of microgravity and Cx43 affect fracture healing outcomes, a novel murine-healing model undergoing different periods of unloading before and during fracture healing, will be developed and characterized. This model will be created by combining the ground-based microgravity analog, hindlimb tail unloading, in conjunction with an established mouse endochondral bone healing model, the stabilized open surgical femoral fracture model. Bone healing outcomes via molecular, histological, mechanical and cellular techniques, will be evaluated in wildtype and Cx43 transgenic mice. Biomarker characterization of healing progression will be evaluated. The outcomes of this research will provide better mechanistic insight into how microgravity and gravitational reloading such as that found during spaceflight and terrestrial exploration, respectively, affects bone healing. Furthermore, this proposal will identify whether possible treatment strategies targeting Cx43, and or other biological targets, is an efficacious approach to augment bone healing during microgravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the capability to address the major NASA Human Research Program (HRP) Gaps in Osteo 1 (Risk of Bone Fracture due to Spaceflight-induced Changes to Bone) and Fracture 1 (We Don't Understand how Spaceflight Affects Fracture Healing). This is the first project to look at mechanical loading interventions and targeted gene deletions using transgenic mice to increase fracture healing during simulated microgravity. If successful, we can combat multiple musculoskeletal health risks (bone, muscle) of spaceflight using a single countermeasure (mechanical loading, Cx43 deletion, and or exercise).
Task Progress:	In Aim 1, we developed a spaceflight-relevant fracture healing model using HLU and found that simulated microgravity leads to significant bone and muscle loss and altered fracture healing, regardless of mouse sex. Our preclinical model combined the ground-based microgravity analog, hindlimb unloading by tail suspension (HLU), in conjunction with an established mouse endochondral bone healing model, the stabilized open surgical femoral fracture model. Aim 1 methods, results, and interpretation have been fully finalized and accepted for publication (PMID: 37001629). We found that continued HLU (simulated microgravity exposure) following fracture led to severely diminished uninjured hindlimb skeletal muscle mass (gastrocnemius and soleus) and femoral bone volume adjacent to the fracture site compared to healthy age-matched controls across mouse sexes. Furthermore, HLU led to significantly decreased periosteal expansion (DPF4) and cartilage and boner tissue formation by DPF14, and trends in increased osteoclastogenesis (DPF14) and decreased woven bone vascular area (DPF14). In contrast, immediate gravitational reloading for 2 weeks after fracture (HLU + R), even following a period of prolonged disuse, was able to increase hindlimb skeletal tissue mass and increase woven bone and cartilage tissue formation, albeit not to healthy control levels, in both mouse sexes. Furthermore, reambulation attenuated osteoclast formation seen in woven bone tissue undergoing disuse. These findings support our Aim 1 hypotheses and mirror the poor/delayed hard and soft callus formation seen in rodent models of spaceflight and the ground based analog hindlimb suspension. Our Aim 2 result demonstrated that constitutive DMP1 Cre Cx43 deficient mice, have noticeable GJA1 recombination in skeletal muscle and bone. We observed no protection with Cx43 deficiency against HLU induced intact contralateral bone loss or improvements in callus mineralization by microCT. However, our results did show protection against HLU induced muscle atrophy based on muscle force generation in vivo. These findings support our Aim 1 hypotheses and mirror the poor/delayed hard and soft callus formation seen in rodent models of spaceflight and the ground based analog hindlimb suspension. Our Aim 2 results using the noninducible DMP1-Cre Cx43 fl/fl mice, which show a lack of a protective effect of DMP1 Cx43 deficiency on bone and fracture healing during HLU refutes our initial hypothesis but does suggest it may have potential benefits on muscle strength. This project supports the collective consensus that musculoskeletal health and bone fracture healing is extremely altered by long-duration microgravity exposure and that gravitational reloading and Cx43 deficiency may alleviate some but not all these of these impairments. These results also have clinical parallels to fracture during extended bedrest and paralysis and suggest that skeletal fracture and altered healing is imminent in any scenario with prolonged unloading. Additional countermeasure testing looking at acute mechanical loading of fractures during simulated microgravity exposure are forthcoming. Additional countermeasure testing looking at partial weight-bearing and galactic cosmic radiation exposure during healing are necessary to understand longterm risks of impaired fracture healing on the Moon or Mars.
Bibliography Type:	Description: (Last Updated: 01/06/2026)
Abstracts for Journals and Proceedings	Buettmann EG, Hoppock GA, Abraham LB, DeNapoli RC, Cornthwaite JR, Denisco JA, Friedman MA, Zhang Y, Windle JJ, McClure MJ, Donahue HJ. "DMP1 lineage Connexin 43 deficiency attenuates decreases in skeletal muscle strength and fracture callus mineralization during hindlimb unloading." 2023 Annual Meeting of the Biomedical Engineering Society, Seattle, WA, October 11-14, 2023. Abstracts. 2023 Annual Meeting of the Biomedical Engineering Society, Seattle, WA, October 11-14, 2023. , Oct-2023
Articles in Peer-reviewed Journals	Zeineddine Y, Friedman MA, Buettmann EG, Abraham LB, Hoppock GA, Donahue HJ. "Genetic diversity modulates the physical and transcriptomic response of skeletal muscle to simulated microgravity in male mice." npj Microgravity. 2023 Dec 1;9:86. https://doi.org/10.1038/s41526-023-00334-8 ; PMID: 38040743; PMCID: PMC10692100 , Dec-2023

Articles in Peer-reviewed Journals	Friedman MA, Buettmann EG, Zeineddine Y, Abraham LB, Hoppock GA, Meas SJ, Zhang Y, Farber CR, Donahue HJ. "Genetic variation influences the skeletal response to hindlimb unloading in the eight founder strains of the diversity outbred mouse population. " J Orthop Res. 2024 Jan;42(1):134-140. https://doi.org/10.1002/jor.25646 ; PubMed PMID: 37321985; PubMed Central PMCID: PMC10721729 , Jan-2024
Articles in Peer-reviewed Journals	Buettmann EG, Yoneda S, Hu P, McKenzie JA, Silva MJ. "Postnatal Osterix but not DMP1 lineage cells significantly contribute to intramembranous ossification in three preclinical models of bone injury." Front Physiol. 2023 Jan 4;13:1083301. https://doi.org/10.3389/fphys.2022.1083301 ; , Jan-2023
Articles in Peer-reviewed Journals	Meas SJ, Daire GM, Buettmann EG, Donahue HJ. "Hindlimb unloading results in accelerated bone loss compared to aging in select bone compartments of female murine femurs." J. Biomech. 2026 Jan;195:113134. https://doi.org/10.1016/j.jbiomech.2025.113134 ; PMID: 41435464 , Jan-2026
Articles in Peer-reviewed Journals	Buettmann EG, Chlebek C, Lockard CA, Clayton SW, Lewis KJ, Collins KH "Post or perish? Social media strategies for disseminating orthopedic research." J Orthop Res. 2023 Aug;41(8):1643-52. https://doi.org/10.1002/jor.25588 ; PMID: 37163368; PMCID: PMC10524931 , Aug-2023
Articles in Peer-reviewed Journals	Buettmann EG, DeNapoli RC, Abraham LB, Denisco JA, Lorenz MR, Friedman MA, Donahue HJ. "Reambulation following hindlimb unloading attenuates disuse-induced changes in murine fracture healing." Bone. 2023 Jul 1;172:116748. https://doi.org/10.1016/j.bone.2023.116748 ; PMID: 37001629 , Jul-2023
Articles in Peer-reviewed Journals	Hoppock GA, Buettmann EG, Denisco JA, Goldscheitter GM, Condyles SN, Juhl 4th OJ, Friedman MA, Zhang Y, Donahue HJ. "Connexin 43 and cell culture substrate differentially regulate OCY454 osteocytic differentiation and signaling to primary bone cells." Am J Physiol Cell Physiol. 2023 Oct 1;325(4):C907-20. https://doi.org/10.1152/ajpcell.00220.2023 ; PMID: 37602413; PMCID: PMC10635658 , Oct-2023
Awards	Buettmann EG. "NIAMS K99 (Grant ID# K99AR082989-01), "Mechanobiology of fracture healing during skeletal disuse", September 2023." Sep-2023
Awards	Buettmann EG. "Orthopedic Research Society Collaborative Exchange Grant, entitled "Investigating Optogenetic Based Approaches to Enhance MSK Regeneration", August 2023." Aug-2023
Awards	Buettmann EG. "Rising Star in BME by Cornell, Johns Hopkins, and Columbia University, September 2023." Sep-2023
Papers from Meeting Proceedings	Buettmann EG, Hoppock GA, Abraham LB, DeNapoli RC, Cornthwaite JR, Denisco JA, Friedman MA, Zhang Y, Windle JJ, McClure MJ, Donahue HJ. "DMP1 lineage Connexin 43 deficiency attenuates decreases in skeletal muscle strength and fracture callus mineralization during hindlimb unloading." 2024 Annual Meeting of the Orthopedic Research Society, Long Beach, California, February 2-6, 2024. Abstracts. 2024 Annual Meeting of the Orthopedic Research Society, Long Beach, California, February 2-6, 2024. , Feb-2024